XINWEN DING

40 St. George Street ◇ Toronto, ON Canada
Homepage ◇ Linkedin ◇ GitHub ◇ Google Scholar

EDUCATION

University of Toronto — *Toronto, Ontario*
Ph.D. in Mathematics — *September 2023 - November 2028 (expected)*

- Advisor: Adam R Stinchcombe
- Thesis: Machine Learning Enhanced Monte Carlo Method for Elliptic Interface Problem
- Relevant coursework: Physics-Informed Neural Representations for Visual Computing, Neural Networks and Deep Learning, Deep Learning Theory & Data Science, Algorithm Design, High Performance Computing, Linear Algebra & Optimization, Mathematical Probability, PDE.

University of Waterloo — *Waterloo, Ontario*
Bachelor of Mathematics — *September 2018 - May 2023*
Triple Honours, Co-operative Program (Major) in:

- Applied Mathematics, Combinatorics & Optimization, and Computational Mathematics
- Major average: 95.21 / 100 ◇ Cumulative average: 94.48 / 100

RESEARCH INTERESTS

Numerical Analysis, Scientific Computing, Scientific Machine Learning

PUBLICATIONS

- Xinwen Ding, Adam R Stinchcombe. **Walk-on-Interfaces: A Monte Carlo Estimator for an Elliptic Interface Problem with Nonhomogeneous Flux Jump Conditions and a Neumann Boundary Condition**. *Journal of Computational Physics, 2026*.
- Xinwen Ding, Christopher Batty. **Differentiable Curl-Noise: Procedural Incompressible Fluid Flows Without Discontinuities**. *ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games*, 2023, Bellevue, WA, USA.

TALKS AND PRESENTATIONS

Conference Presentation

- Walk-on-Interfaces: A Monte Carlo Estimator for Elliptic Interface Problem, *41st Southern Ontario Numerical Analysis Day, 2026*
- Differentiable Curl-Noise: Procedural Incompressible Fluid Flows Without Discontinuities, *ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games, 2023*

Poster Presentation

- Walk-on-Interfaces: A Monte Carlo Estimator for Elliptic Interface Problem, *Canadian Mathematical Society (CMS) Winter Meeting, 2025*

HONOURS AND AWARDS

NSERC Canada Graduate Research Scholarship - Doctoral (CGRS D) ($120,000) — *2026 - 2029*

- A prestigious national award reserved for top-ranked doctoral students in Canada. Successfully competed within a strict 15% international student quota based on academic excellence, research potential, and leadership.

Martin Shubik Graduate Award in Mathematics ($5,000) *2024 - 2025*
Connaught International Scholarship ($50,000) *2023 - 2028*

- A cost-sharing initiative designed to assist graduate units in recruiting and supporting top international students by providing a top-up scholarship in addition to the standard funding package.

Jessie W.H. Zou Memorial Award (Honours Mention) *2023*

- Presented annually to an undergraduate student enrolled in his or her final year of any program within the Faculty of Mathematics. Students must have demonstrated excellence in research and have been nominated by a faculty member who has supervised that research.

President's Research Award ($1,500 each year) *2022-23, 2021-22 Academic Years*

- Awards to undergraduate students who are undertaking a full- or part-time research experience, under the supervision of a University of Waterloo researcher.

Mathematics Undergraduate Research Award ($6,000) *2022-23 Academic Year*
Term Dean's Honours List *Fall 2018 - Present*
R.A. Wentzell Memorial Scholarship ($1,000) *2020-21 Academic Year*

- Awards to the top male and top female students who are enrolled in year three in the Department of Applied Mathematics on the basis of academic excellence.

University of Waterloo Faculty of Mathematics Scholarship ($5,000) *2018-19 Academic Year*
University of Waterloo President's Scholarship of Distinction ($2,000) *2018-19 Academic Year*

RESEARCH EXPERIENCE

Learning-based Monte Carlo Method for Elliptical Interface Problem
Department of Mathematics - University of Toronto *May 2024 - September 2025*
Supervisor: Prof. Adam R Stinchcombe

- Proposed and implemented a novel Monte Carlo method to solve Neumann elliptic interface problems.
- Integrated a neural network in PyTorch to represent continuous solutions across irregular geometries.
- Validated method performance on challenging test cases; first-authored manuscript submitted.

Musical Signal Decomposition Via Non-negative Matrix Factorization (NMF)
Department of Combinatorics and Optimization, and *September 2022 - April 2023*
Department of Applied Mathematics - University of Waterloo *Undergraduate Research Assistant*
Supervisors: Prof. Stephen Vavasis, Prof. Giang Tran, Prof. Andersen Ang

- Formulated an NMF-based framework for decomposing dense spectrograms of musical audio.
- Designed optimization-based models enabling NMF to extract and cluster features of musical signals.
- Tested the model with real recordings; Achieved successful decomposition of musical signals (e.g., *Für Elise Mary Had a Little Lamb*) into distinct instrument and note bases.

Differentiable Curl-Noise: A Procedural Visualization of Incompressible Fluid Flow
Cheriton School of Computer Science - University of Waterloo *September 2021 - May 2023*
Supervisor: Prof. Christopher Batty *Undergraduate Research Assistant*

- Constructed a C^1 procedural method to simulate incompressible fluid and proved its differentiability; Smoothed the fluid's potential and velocity field, which enables isocontour error correction.
- Animated fluid flow using C++ and OpenGL; Pushing the fluid to float past arbitrarily shaped objects that are placed at arbitrary locations, under the effect of a C^1 potential field.
- First-authored paper accepted by *ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games*, 2023. To appear in the journal *"Proceedings of the ACM on Computer Graphics and Interactive Techniques" (PACMCGIT).*

Creating Large and Uniform Optical Traps Arrays Using Spatial Light Modulators (SLM)
Institute for Quantum Computing — *January 2021 - August 2021*
Supervisor: Dr. Alexandre Cooper-Roy, Prof. David Cory — *Full-time Co-op*

- Developed the lab's first real-time, experiment-interfaceable Python simulation package; Simulated laser beam, laser beam propagation, aberration, lens focusing effect, and all experiment devices.
- Optimized a solution to a phase retrieval (inverse) problem; Retrieved phase for large ($\mathcal{O}(10^3)$ traps) and uniform (10^{-8} non-uniformity) trap arrays in 0.4 s, 600 times faster than the original solution.
- Developed an iterative algorithm to perform accurate, software-independent calibration for SLM; Reduced the measurement error (metric: ℓ_2-norm) to $\frac{1}{10}$ of the error under commercial solution.

PROJECT EXPERIENCE

Summer Geometry Initiative *(SGI)* — *July 2022 - August 2022*
Massachusetts Institute of Technology — *2022 SGI Fellow*

- Attended hands-on tutorials introducing the theory and practice of geometry processing; Participated in multiple research projects, and wrote SGI blog posts: *[tutorial week]*, *[Project 1]*, *[Project 2]*.
- **Project 1: Plane and Edge Detection in Point Clouds**
  Mentor: Prof. Yusuf Sahillioglu
  · Implemented the RANSAC algorithm to detect dominant planes and edges from a given point cloud.
- **Project 2: Neural Implicit Boundary Representations**
  Mentor: Benjamin Jones, Prof. Adriana Schulz
  · Explored and implemented a two-phase method to define a continuous relaxation of CAD geometry.

WORK EXPERIENCE

Ontario Ministry of Infrastructure — *September 2019 - December 2019*
Research Analyst — *Toronto, ON*
Supervisor: Michael Leigh, Roy Hulli — *Full-time Co-op*

- Researched on the infrastructure expenditure deficit problem; Explored and collected data; Standardized, and analyzed provincial expenditure via SQL.
- Designed and created a dynamic dashboard independently using VBA and SQL to visualize 6000 infrastructure project records (a prototyping version of the current *Ontario Builds* dashboard).
- Performed data modelling for transportation and correction sector for Ontario Multi-year Plan; Calculated financial investment data from existing government records that fits the budget model.

TEACHING

University of Toronto — *Department of Mathematics*

Course Instructor

- MAT 244 (UTSG) - Introduction to Ordinary Differential Equations — *Summer 2026*

Graduate Teaching Assistant

- APM 462 (UTSG) - Nonlinear Optimization — *Winter 2026*
- APM 236 (UTSG) - Applications of Linear Programming — *Fall 2025*
- MAT244 (UTSG) - Introduction to Ordinary Differential Equations — *Fall 2024 & Winter 2025*
- MAT133Y1Y (UTSG) - Calculus and Linear Algebra for Commerce — *Fall 2023 & Winter 2024*
- Math Learning Center — *Winter 2025 & Winter 2026*

University of Waterloo — *Faculty of Mathematics*

Undergraduate Teaching Assistant (Math Tutorial Center)

- MATH 235 - Linear Algebra 2 for Honours Mathematics — *Fall 2021*
- MATH 237 - Calculus 3 for Honours Mathematics — *Winter 2022*

SERVICE AND OUTREACH

Volunteer Mentor & Presenter, Pursue STEM *April 2026*

- Facilitated academic guidance for a University of Toronto outreach initiative designed to support and encourage Black high school students pursuing STEM study.
- Led interactive workshop sessions and delivered a guest presentation titled "Parabolas as Envelopes of Lines" to bridge high school algebra with advanced geometric concepts.

Math Graduate Students' Association (MGSA)

- Course Planning Committe (Member) *2024 - 2025*
- Graduate Student Union (GSU) representative *2023 - 2024*

SKILLS

Programming Languages	Python (PyTorch, TensorFlow), MATLAB, C/C++
Tools	MPI, Git, Linux, OpenGL, Blender, LaTeX
Research Areas	Scientific computing, scientific machine learning, Monte Carlo methods data-driven modeling, optimization